Cognitive Load Mapper

Stats 418 Project Proposal Presentation

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Overview

Cognitive Load Mapper: Understanding cognitive load in reading

Problem	Long-form articles and other text can have varying cognitive load based on how “easy” or “hard” they are to read.
Goal	Understand how the cognitive load for the reader changes paragraph-by-paragraph and influences reader engagement.
Target Users	Anyone who wants to examine the cognitive complexity of their writing or understand where a piece of text becomes cognitively demanding.

Data Collection

Data: Substack web scraping via BeautifulSoup and requests

Current Data Collection Pipeline

```
# data_collector.py — sitemap → HTML → JSON
for newsletter, base_url in
TARGET_NEWSLETTERS:
  urls = get_sitemap_urls(base_url)
  post = scrape_post(url, newsletter)
  # → raw_posts.json + paragraphs.csv

# paragraph_segmenter.py — NLTK/spaCy SBD
# → segmented_paragraphs.csv + sentences.csv
```

Data Collection Progress

104 posts, 5034 paragraphs

```
# raw_posts.json → raw data json
[ { "post_id": str, "newsletter": str, "url": str, "title":
str, "publish_date": str, "full_body_text": str,
"paragraphs": [ str, ... ], "paragraph_count": int,
"word_count": int, "estimated_read_time": int, "like_count":
int, "comment_count": int, "scraped_at": str }]
```

Notable Challenges

Rate limiting: 2.5 s delay between requests

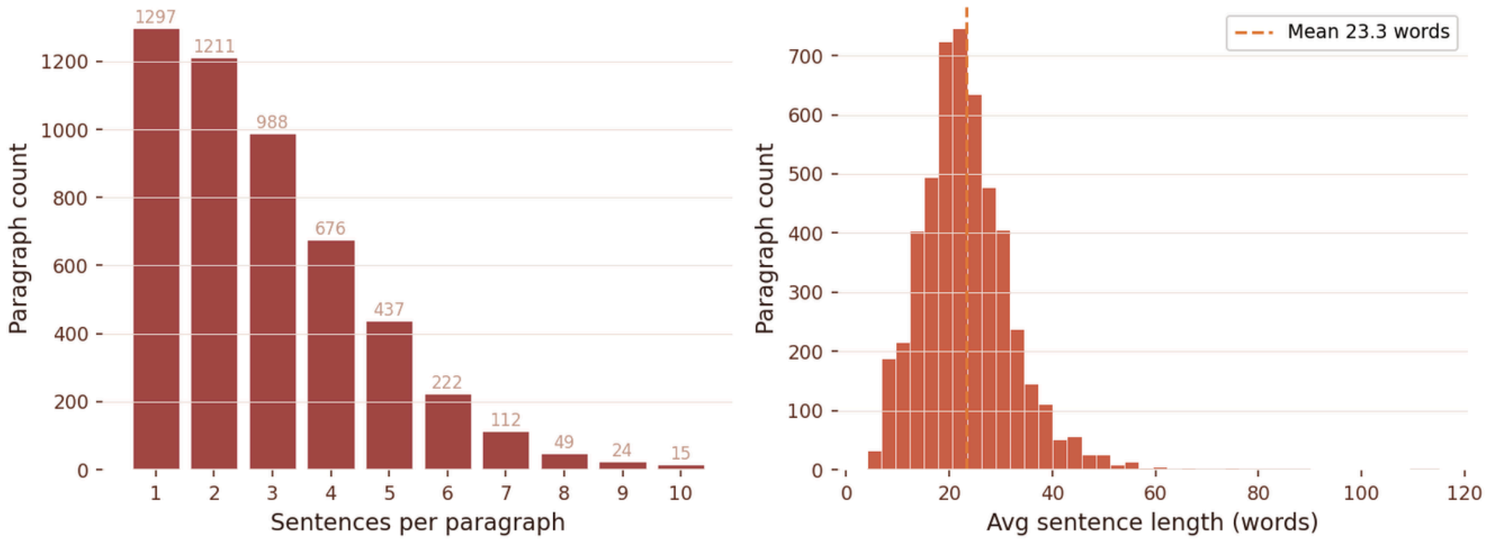
Limited engagement data: Likes and comment counts are partially inaccessible without authentication on some newsletters

Limited posts: Collected fewer posts than target (reached 5000+ paragraph target before reaching 500+ post target)

EDA

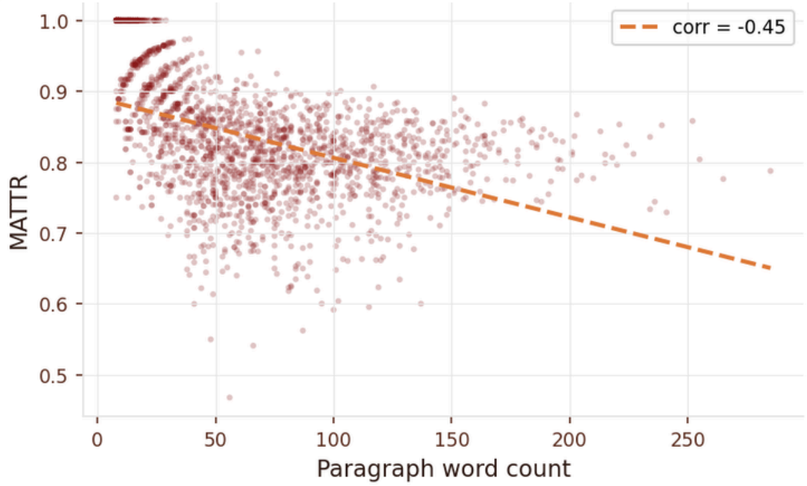
104 posts, **5034** paragraphs

Paragraph/Sentence Statistics



Most paragraphs are **1-3 sentences**. Average sentence length is **23.3 words**.

Lexical Diversity



Most paragraphs have MATTR values around **0.75–0.90**, which suggests that fairly varied vocabulary is being used.

Modeling

Target Variable(s)

Primary target: engagement depth score, based on normalized likes/comments when available

Secondary target: unsupervised gradient shape label if engagement data is too sparse

Potential Feature Set

Readability: fk_grade, smog_index, dale_chall_score, ari, flesch_reading_ease

Lexical: mattr, avg_syllables_per_word

Discourse: inferential_gap (cosine dist. between consecutive paragraph embeddings)

Entities: new_entity_rate, cumulative_entity_count

Presupposition: trigger_density (still, even, already, realize…)

Sentiment: NRC emotion rates

Model Plan

Tier	Model	Justification	Target	Evaluation
Baseline	Linear regression on mean FK grade only	Tests whether a single readability score predicts engagement	Engagement score	RMSE on held-out 20% (stratified by newsletter)
Main	PCA (10 features) → XGBoost	Strong for tabular features, works with nonlinear relationships	Shape class	F1-macro across 5 classes

Current Progress

- data_collector.py
- paragraph_segmenter.py
- feature_engineering.py

- 5,034 paragraphs ✓
- 11 feature groups ✓

Next Steps

Week 6
Complete data collection (105 → 500+ posts, 1 → 6 newsletters)

Week 7
Complete EDA and feature engineering

Week 8
Train and evaluate models

Week 9
Improve models by iterating on feature engineering/analysis/ algorithms

Week 10
Build and deploy Streamlit interface

Risks & Mitigation

Missing likes/comments → Use publicly available counts only; otherwise focus on unsupervised gradient labels

Model signal may be weak and indirect → Final result as a writing diagnostic tool, not a perfect engagement predictor